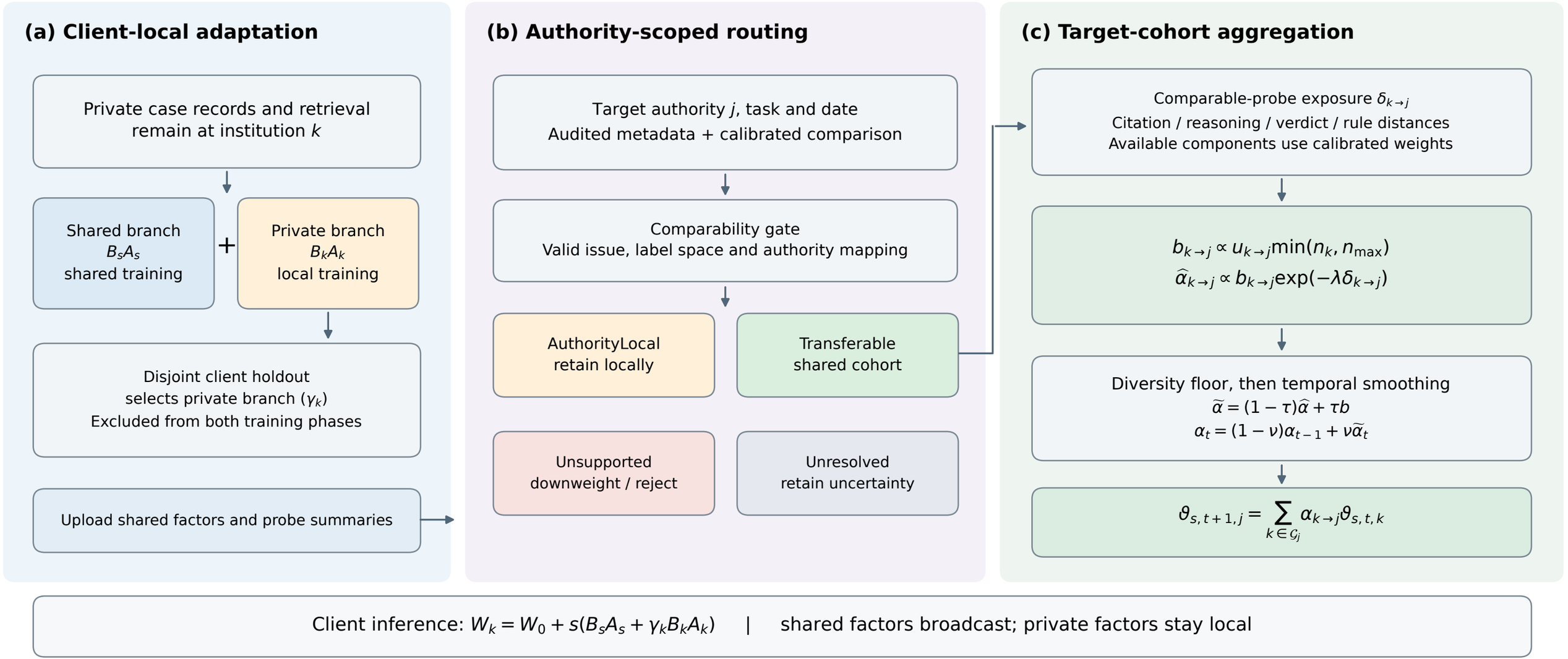
(a) Client-local adaptation
Private case records and retrieval remain at institution k
Shared branch B_sA_s shared training
+
Private branch B_kA_k local training
Disjoint client holdout selects private branch (γ_k) Excluded from both training phases
Upload shared factors and probe summaries
(b) Authority-scoped routing
Target authority j, task and date Audited metadata + calibrated comparison
Comparability gate Valid issue, label space and authority mapping
AuthorityLocal retain locally
Transferable shared cohort
Unsupported downweight / reject
Unresolved retain uncertainty
(c) Target-cohort aggregation
Comparable-probe exposure $\delta_{k\to j}$
Citation / reasoning / verdict / rule distances
Available components use calibrated weights
$b_{k\to j} \propto u_{k\to j}\min(n_k, n_{\max})$
$\widehat{\alpha}_{k\to j} \propto b_{k\to j}\exp(-\lambda\delta_{k\to j})$
Diversity floor, then temporal smoothing
$\widetilde{\alpha} = (1-\tau)\widehat{\alpha} + \tau b$
$\alpha_t = (1-\nu)\alpha_{t-1} + \nu\widetilde{\alpha}_t$
$\vartheta_{s,t+1,j} = \sum_{k\in\mathcal{C}_j} \alpha_{k\to j}\vartheta_{s,t,k}$
Client inference: $W_k = W_0 + s(B_sA_s + \gamma_kB_kA_k)$ | shared factors broadcast; private factors stay local